# Modeling 8-port maze behavior

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## 1 POMDP model

We will provide a POMDP model of the mouse poke behavior as follows. In this model, the discrete time underlying the Markov chain is the poke index which we denote by $t$, and we denote the poke sequence by $p_t$, where at each time step $p_t \in \{1, \dots, 8\}$. Several different pokes can occur in the same trial, and we use $\ell(t) \in \mathbb{N}$ to denote the trial in which the $t$-th poke happened. Currently, we do not model the agent's decision making regarding how many time to poke (or whether or not to continue poking) in a given trial. Basically, we assume in each trial the agent is forced to poke a pre-specified number of pokes, which number can vary across trials.

We will let the internal model of the mouse to deviate from the ideal observer model that represents the experimental design faithfully.

**Markov chain:** The underlying Markov chain (MC) is trivial in the sense that there are no transitions (the transition matrix is the identity matrix). The are two discrete state variables: $r$ the reward port, and $\tau$ the trial number in which reward becomes available. $r$ has eight possibilities and $\tau$ has $T$ possibility (where $T$ is about 20). So the total number of MC states is $8T$.

**Prior distribution/beliefs:** According to the experimental design, the prior probability of the port in the recall/afternoon session should be a delta function at the morning session. But we will allow the mouse to have imperfect memory and will assume a finite-width/uncertainty distribution: e.g. a von Misses centered at the correct reward port, or a mixture of a delta function and the uniform distribution. We denote this prior distribution of the mouse by $f_\theta(r)$ where $\theta$ are some parameters (e.g. the width of the VM or the mixture weight). The idea is that we will fit this model to the behavioural data and infer those parameters (among others) to make statements about memory and how it depends on various things such as pharmacological manipulations.

The distribution of $T$ is uniform and for now we will assume that the mouse knows this. But more generally we can assume this is a parametrized distribution $g_\psi(\tau)$ with parameters $\psi$.

We assume $r$ and $\tau$ are a priori independent in the mouse's head, and so

$$\mathbb{P}(r, \tau) = f_\theta(r) g_\psi(\tau). \tag{1}$$ {prior}

**Observation distributions:** The observation at $t$-th poke, $p_t$, is a binary variable, $w_t$ indicating whether or not water was received.

According to the experimental design, the observation distributions is as follows:

$$\mathbb{P}(w_t = 1 |\, p_t, r, \tau) = 1(p_t = r)1(\ell(t) \geq \tau) \tag{2}$$

where 1(statement) is one if statement is true and zero otherwise.

Again we can allow the mouse's internal observation model to deviate from this 0/1 model, e.g. we can allow the mouse to assume that even non-reward ports have a small probability of giving reward, or that even before trial $\tau$ there is a small probability of receiving reward. But for now we won't do that.

**Actions:** the actions of the mouse are nothing but the poke choices $p_t$.

**Reward function:** The reward variables are the same as the observation variables $r_t = w_t$ and the total future return at time $t$ is given by $G_t = w_t + w_{t+1} + \ldots$.

## 2 Bayesian Belief updates

We denote the agent's beliefs just after the $t$-th poke, *i.e.*, the conditional distribution of states conditioned on the observations (and actions) up to (and including) time $t$, by $b_t(r, \tau) = \mathbb{P}(r, \tau | \{w_{t'}\}_{t'=1}^{t}, \{p_{t'}\}_{t'=1}^{t})$. These are proportional to the joint distribution of the Markov states after the $t$-th poke, conditioned on the poke choices up to that point, *i.e.*,

$$b_t(r,\tau) \propto \mathbb{P}(r, \tau, \{w_{t'}\}_{t'=1}^{t} | \{p_{t'}\}_{t'=1}^{t}) = f_\theta(r) g_\psi(\tau) \prod_{t'=1}^{t} [1(p_{t'} = r)1(\ell(t') \geq \tau)]^{w_{t'}} \; [1 - 1(p_{t'} = r)1(\ell(t') \geq \tau)]^{1-w_{t'}} \tag{3}$$

{eq:beliefs

(to get the beliefs, after calculating the r.h.s. you have to normalize it so its sum over all possible values of $r, \tau$ is 1). From this we can write the belief update from poke to poke

$$b_t(r,\tau) \;\propto\; b_{t-1}(r,\tau) \;\; [1(p_t = r)1(\ell(t) \geq \tau)]^{w_t} \; [1 - 1(p_t = r)1(\ell(t) \geq \tau)]^{1-w_t} \tag{4}$$

{eq:belief-

Note that in our notation $b_0(r, \tau)$ is the prior belief, given by Eq. (1).

We can also allow for a non-ideal-observer water-likelihood function, according to which mice may potentially assume a potentially nonzero (albeit small) probability, $\rho$, for getting reward from non-reward ports, and a potentially nonzero probability, $\sigma$, for getting reward in water-unavailable trials. Defining the function

$$1_\lambda(\text{condition}) = 1(\text{condition}) + \lambda(1 - 1(\text{condition})) = \lambda + (1 - \lambda)1(\text{condition}) \tag{5}$$

$$1 - 1_\lambda(\text{condition}) = (1 - \lambda)(1 - 1(\text{condition})) \tag{6}$$

the generalized form of Eq. (4) becomes

$$b_t(r,\tau) \;\propto\; b_{t-1}(r,\tau) \;\; [1_\rho(p_t = r)1_\sigma(\ell(t) \geq \tau)]^{w_t} \; [1 - 1_\rho(p_t = r)1_\sigma(\ell(t) \geq \tau)]^{1-w_t} \tag{7}$$

{eq:belief-

(with Eq. (4) corresponding to the case $\rho = \sigma = 0$).

### 2.1 Limiting to all trials before (and including) the first water

Once they mice find and recieve water from the reward port, stereotypically (but with some exceptions across mice or across trials) in all subsequent trials they just go straight to the reward port and only poke that. So the interesting behaviour happens in preceding trials.

We will thus formulate the MDP as one were there is no reward (or we can assume a constant negative reward of -1) in all trials without water and we consider the first water trial as a **terminal state** (which yields positive or even zero reward) in the language of RL, upon reaching which the episode is terminated. In this view the aim of the mouse is to reach this terminal state as soon as possible.

Henceworth we will consider this case only. In this case in all relevant trials[1] no water was received, i.e. $w_t = 0$. So Eqs. (3)–(4) can be simplified to

$$b_t(r,\tau) \propto f_\theta(r) g_\psi(\tau) \prod_{t'=1}^{t} [1 - 1_\rho(p_{t'} = r) 1_\sigma(\ell(t') \geq \tau)] \tag{8}$$ {eq:beliefs

and

$$b_t(r,\tau) \propto b_{t-1}(r,\tau) \; [1 - 1_\rho(p_t = r) 1_\sigma(\ell(t) \geq \tau)] \tag{9}$$ {eq:belief

#### 2.1.1 Last visit variables

In the case $\rho = \sigma = 0$, a simplified representation of belief state offers itself. In Eq. (8), let us divide the time-steps in the product $\prod_{1:t}$ into 8 groups depending on which port was poked. Denoting the subset of time-indices up to time $t$ in which the $r$-th port was poked by $S_t(r) := \{t' \,|\, t' \leq t \wedge p_{t'} = r\}$, we can see that only the subset $S_t(r)$ contribute a non-trivial factor (other than 1) to the update for $b_t(r,\tau)$:

$$b_t(r,\tau) \propto f_\theta(r) g_\psi(\tau) \prod_{r'=1}^{8} \prod_{t' \in S_t(r')} [1 - 1(r' = r) 1(\tau \leq \ell(t'))] \tag{10}$$

$$= f_\theta(r) g_\psi(\tau) \prod_{t' \in S_t(r)} [1 - 1(\tau \leq \ell(t'))] \tag{11}$$

$$= f_\theta(r) g_\psi(\tau) \prod_{t' \in S_t(r)} 1(\tau > \ell(t')) \tag{12}$$ {beliefs so

(again, this is assuming no water was obtained up to and including the $t$-th poke). On the right side of Eq. (12), each factor $1(\tau > \ell(t'))$ simply zeroes out all entries in the $r$-th row of the belief matrix before and including column $\ell(t')$. Thus the product $\prod_{t' \in S_t(r)} 1(\tau > \ell(t'))$ would zero out all belief elements up to an including the last trial —up to and including the trial in which poke $t$ occurred— in which the $r$-th port was poked. Denoting this trial by

$$T_r^t = \max \ell(S_t(r)) = \max\{\ell(t') \,|\, t' \leq t \wedge p_{t'} = r\} \tag{13}$$
$$= \text{Most recent trial, just after the } t\text{-th poke, in which port } r \text{ was poked}$$

we can thus rewrite Eq. (12) as

$$b_t(r,\tau) \propto f_\theta(r) g_\psi(\tau) \; 1(\tau > T_r^t). \tag{14}$$ {eq:beliefs

[1] Since once termination is reached no further actions are taken by the agent, we don't need to consider the updated belief after water has finally been received upon reaching the termination state.

Note that $0 \le T_r^t \le \ell(t)$: we have $T_r^t = 0$ when port $r$ has never been poked so far (including at time $t$), and $T_r^t = \ell(t)$ iff $p_{t'} = r$ for some poke $t' \le t$ in trial $\ell(t)$ (*i.e.*, iff $r$ was poked in trial $\ell(t)$, before or at $t$).

The last visit representation is particularly useful in the case where the experimental probability of $\tau$ (or really the mouse internal model of it), $g_\psi$, is a uniform distribution.

#### 2.1.2 Last visit representation for nonzero $\rho$ or $\sigma$

Let us rewrite the derivaiton Eq. (12) for nonzero $\rho$ or $\sigma$

$$b_t(r,\tau) \propto f_\theta(r) g_\psi(\tau) \prod_{r'=1}^{8} \prod_{t' \in S_t(r')} [1 - 1_\rho(r' = r) 1_\sigma(\tau \le \ell(t'))] \tag{15}$$

$$= f_\theta(r) g_\psi(\tau) \prod_{t' \in S_t(r)} [1 - 1_\sigma(\tau \le \ell(t'))] \prod_{t' \notin S_t(r)} [1 - \rho 1_\sigma(\tau \le \ell(t'))] \tag{16}$$

$$= f_\theta(r) g_\psi(\tau) (1-\sigma)^{n_t^r} \prod_{t' \in S_t(r)} [1 - 1(\tau \le \ell(t'))] \prod_{t' \notin S_t(r)} [1 - \rho 1_\sigma(\tau \le \ell(t'))] \tag{17}$$

$$= f_\theta(r) g_\psi(\tau) (1-\sigma)^{n_t^r} \prod_{t' \in S_t(r)} 1(\tau > \ell(t')) \prod_{t' \notin S_t(r)} [1 - \rho 1_\sigma(\tau \le \ell(t'))] \tag{18}$$

$$= f_\theta(r) g_\psi(\tau) 1(\tau > T_r^t) (1-\sigma)^{n_t^r} \prod_{t' \notin S_t(r)} [1 - \rho 1_\sigma(\tau \le \ell(t'))] \tag{19}$$

{beliefs so

where $n_t^r = |S_t(r)|$ is the number of times port $r$ has been poked at or before $t$. The factor $1 - \rho 1_\sigma(\tau \le \ell(t'))$ can be rewritten as $1 - \rho + \rho(1 - 1_\sigma(\tau \le \ell(t'))) = 1 - \rho + \rho(1-\sigma)(1 - 1(\tau \le \ell(t'))) = 1 - \rho + \rho(1-\sigma)1(\tau > \ell(t'))$ which is $1 - \rho\sigma$ when $\tau > \ell(t')$ and $1 - \rho$ otherwise. We thus obtain

$$b_t(r,\tau) \propto f_\theta(r) g_\psi(\tau) 1(\tau > T_r^t) (1-\sigma)^{n_t^r} \prod_{t' \notin S_t(r), \ell(t') < \tau} (1 - \rho\sigma) \prod_{t' \notin S_t(r), \ell(t') \ge \tau} (1 - \rho) \tag{20}$$

or

$$b_t(r,\tau) \propto f_\theta(r) g_\psi(\tau)\ 1(\tau > T_r^t) (1-\sigma)^{n_t^r} (1 - \rho\sigma)^{n_t(r,\tau)} (1-\rho)^{t - n_t^r - n_t(r,\tau)}. \tag{21}$$

{eq:beliefs

where $n_t(r,\tau)$ is the cardinality of the set $\{t' | t' \le t, t' \notin S_t(r), \ell(t') < \tau\}$ which is $t - n_t^r$ when $\ell(t) < \tau$ (because then all $t'$ in the product satisfy $\ell(t') < \tau$, so we can drop that condition) and $n_t(r,\tau) = t_\tau - n_{t_\tau}^r$, otherwise, where $t_\tau$ is the total number of pokes (or time steps) *before* the start of trial $\tau$.

### 2.2 Reduction to MDP and simplified belief-state representation

Any POMDP can be reduced to a (fully observed) MDP with the (objective) states replaced by beliefs. What needs to be done is calculating the transition probability between belief-states. In our case, the transitions to non-terminal states are deterministic and are given by Eq. (9). I.e. (conditioned on the new action yielding no water) the update only depends on the action taken, $p_{t+1}$. However, there is a stochastic probability, or rather belief, to get water, i.e. to enter the terminal state (which we will calculate below).

**Simplified belief-state representation:** By inspecting Eq. (8), we see that the (dynamic aspect of the) beliefs are determined by and thus information theoretically equivalent to, the sequence of

preceding actions $p_{1:t}$. So we can denote the state, $s_t$, of the (POMDP-derived belief-based) MDP, equivalently by $p_{1:t}$, instead of by $b_t$; with $p_{1:0}$ (the empty sequence) corresponding to $b_0$, *i.e.*, the prior beliefs. Interestingly the $p_t$ thus play a dual role: they are both actions and states in this MDP. More precisely,

$$s_t = \begin{cases} p_{1:t} \\ \text{or} \\ \text{TS} \end{cases} \tag{22}$$

$$a_t = p_{t+1} \tag{23}$$

where TS denotes the (special) terminal state. The transition probabilities of this MDP's Markov chain, i.e. the probability of ending in state $s_{t+1}$, conditioned on starting in $s_t$ and taking action $a_t$, are given by

$$(s_t, a_t) = (p_{1:t}, p_{t+1}) \rightarrow \begin{cases} s_{t+1} = p_{1:t+1} & \text{with prob. } 1 - \mathcal{T}_{t+1} \\ \\ s_{t+1} = \text{TS} & \text{with prob. } \mathcal{T}_{t+1} \end{cases} \tag{24}$$

where the termination probability, $\mathcal{T}_t := \mathcal{T}_t(p_{1:t})$, is the *subjective belief* that the $t$-th poke will yield water

$$\begin{aligned} \text{Termination probability} = \mathbb{P}(w_t = 1 | b_{t-1}, p_t) &= \sum_{r,\tau} \mathbb{P}_t(w_t = 1 | r, \tau, p_t)\ b_{t-1}(r, \tau) \\ &= \sum_{r,\tau} 1(\tau \leq \ell(t)) 1(p_t = r)\ b_{t-1}(r, \tau) \end{aligned} \tag{25}$$

or

$$\mathcal{T}_t := \text{Termination probability} = \sum_{\tau \leq \ell(t)} b_{t-1}(r = p_t, \tau) \tag{26}$$

{eq:termina

In the case of a uniform prior $\tau$-distribution, $g_\psi$, from Eq. (26) and Eq. (14), we see that the termination probability can be written as

$$\mathcal{T}_t = \mathcal{T}_t(p_{1:t}) = \sum_{\tau \leq \ell(t)} b_{t-1}(p_t, \tau) \tag{27}$$

$$= \frac{f_\theta(p_t) \sum_{\tau \leq \ell(t)} g_\psi(\tau)\ 1(\tau > T_{p_t}^{t-1})}{\sum_{r=1}^{8} \sum_{\tau=1}^{T} f_\theta(r) g_\psi(\tau)\ 1(\tau > T_r^{t-1})} \tag{28}$$

$$= \frac{f_\theta(p_t) \sum_{\tau \leq \ell(t)}\ 1(\tau > T_{p_t}^{t-1})}{\sum_{r=1}^{8} \sum_{\tau=1}^{T} f_\theta(r)\ 1(\tau > T_r^{t-1})} \tag{29}$$

or

$$\mathcal{T}_t(p_{1:t}) = \frac{f_\theta(p_t)(\ell(t) - T_{p_t}^{t-1})}{\sum_{r=1}^{8} f_\theta(r)(T - T_r^{t-1})} \tag{30}$$

{eq:termina

Let us go back to Eq. (14) and calculate the the marginal posterior probability (i.e. the marginal belief) of the reward port. We denote this marginal belief, just after poke $t$, by $m_t(r) := m_t(r \,|\, p_{1:t})$. To calculate it, we have to integerate out $\tau$:

$$m_t(r) = \sum_{\tau=1}^{T} b_t(r, t) = \frac{f_\theta(r) \sum_{\tau=1}^{T} g_\psi(\tau)\ 1(\tau > T_{p=r}^{t})}{\sum_{r'=1}^{8} \sum_{\tau=1}^{T} f_\theta(r') g_\psi(\tau)\ 1(\tau > T_{p=r'}^{t})} \tag{31}$$

$$= \frac{f_\theta(r) \sum_{\tau=1}^{T}\ 1(\tau > T_{p=r}^{t})}{\sum_{r'=1}^{8} f_\theta(r') \sum_{\tau=1}^{T}\ 1(\tau > T_{p=r'}^{t})} \tag{32}$$

or

$$m_t(r) := m_t(r|p_{1:t}) = \frac{f_\theta(r)(T - T_r^t)}{\sum_{r'=1}^{8} f_\theta(r')(T - T_{r'}^t)}. \tag{33}$$ {eq:marginal

Comparing this with Eq. (30), we can express the termination probability also as

$$\mathcal{T}_t(p_{1:t}) = m_{t-1}(p_t)\ \frac{\ell(t) - T_{p_t}^{t-1}}{T - T_{p_t}^{t-1}} \tag{34}$$ {eq:termina

Based on the row-zeroing story given after Eq. (12), for uniform $g_\psi$, the conditional belief over $\tau$ at time $t$, conditioned on reward port being $r$, is the uniform distribution over the trial indices $T_r^t + 1, \ldots, T$. Thus, by the product rule of probabilities[2], the joint belief can also be represented as

$$b_t(r, \tau) = m_t(r)\ \frac{1(\tau > T_r^t)}{T - T_r^t}. \tag{35}$$

Thus the fraction on the rhs of Eq. (34) is nothing but the conditional posterior probability/belief, at time $t-1$, that $\tau$ happened before or at the current trial: i.e. that reward is now available. So Eq. (34) makes perfect sense intuitively: the (subjective) probability of termination, i.e. the probability of finally receiving water at poke $p_t$, is exactly the (subjective) posterior probability, $m_{t-1}(r = p_t)$, that I am now poking the correct port, times the subjective conditional probability/belief that $\tau$ is the a past trial or the current one and thus water is now flowing.

### 2.3 Greedy policy

A reasonable greedy policy is one where, at every step, chooses the action that maximizes the expected immediate reward, which (see next section) is given by the termination probability $\mathcal{T}_t$. So using Eq. (34)

$$p_t = a_{t-1} = \pi(s_{t-1} = p_{1:t-1}) = \arg\max_p \mathcal{T}_t(p_{1:t-1}, p_t = p) \tag{36}$$ {greedy pol

$$= \arg\max_p f_\theta(p)(\ell(t) - T_p^{t-1}). \tag{37}$$

Alternatively, at every trial, the mice can choose the port that has the highest posterior probability of being the reward port according to the mouse's current beliefs. In other words

$$p_t = a_{t-1} = \pi(s_{t-1} = p_{1:t-1}) = \arg\max_p m_{t-1}(p \,|\, p_{1:t-1}) \tag{38}$$ {max belief

$$= \arg\max_p f_\theta(p)(T - T_p^{t-1}) \tag{39}$$

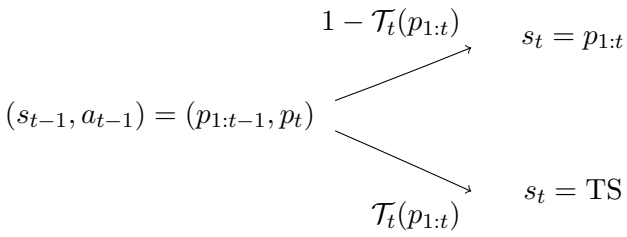
where we used Eq. (33) to write the 2nd line.

This second, maximal-belief-port policy, Eq. (38), is presumably sub-optimal relative to the greedy policy Eq. (36), because it does not take into account beliefs over availability of water. As an extreme case consider the state where at the beginning of a trial some port (perhaps the one with maximum marginal belief or whatever) was poked, without yielding water, with trailing pokes remaining in that trial. The agent reasons that either water is not yet available in this trial,

[2] That $P(r, \tau) = P(\tau|r)P(r)$.

including for its remaining pokes, or water is available but the port I have poked is not the reward port. Given I still *have to* poke (remember we are assuming the agent does not have free will over how many pokes to make in each trial), the agent reasons, I may as well try a port other than the one(s) I have already poked in this trial then, in case reward is available, but the port I poked first is not the reward port. This is what the first policy does, but not the second policy, which in general may keep poking a port that did not yield reward more than once in the same trial; the first policy never pokes a port twice in the same trial (up to an including the first rewarded poke).

## 3 Bellman optimality equation




The Bellman optimiality equation for the state-action value function

$$Q(s_t, a_t) = r(s_t, a_t) + \gamma \mathbb{E}[V(S_{t+1})|s_t, a_t] \tag{40}$$ {eq:bellman

$$V(s) = \max_a Q(s, a). \tag{41}$$ {eq:bellman

where $r$ is the expected immediate reward. In words, the first equation says the value of choosing action $a$ in state $s$ is the immediate reward plus the discounted value of the state you end up in; the second equatino says, for the optimal policy, the state value, $V(s)$, is equal to the $Q$ for the best action.

For $Q(s_t, a_t) = Q(p_{1:t-1}, p_t)$ and the state you end up with and the corresponding probabilities are given by the diagram above. The immediate reward is 1 if you end up in TS and zero otherwise. So the expected reward is given by the termination probability $\mathcal{T}_t$. The value of the TS is zero, so the second term in the bellman equation would be nonzero only if we do not end up in the TS, and therefore is proportional to $1 - \mathcal{T}_t$. Thus for non-terminal states we obtain

$$Q(p_{1:t-1}, a = p_t) = \mathcal{T}(p_{1:t}) + \gamma(1 - \mathcal{T}(p_{1:t})) \max_a Q(p_{1:t}, a). \tag{42}$$ {bellman Q}

while $Q(\text{TS}, a) = 0$. By Eq. (41), the corresponding equation for the state value function is

$$V(p_{1:t-1}) = \max_{p_t} \left[ \mathcal{T}(p_{1:t}) + \gamma(1 - \mathcal{T}(p_{1:t}))V(p_{1:t}) \right]. \tag{43}$$ {eq:bellman

## 4 Log-Likelihood Evaluation of Policy Models

To evaluate how well a trained policy matches experimental behavior, we compute the log-likelihood of the observed poke sequences under the model. The procedure is summarized below for both reinforcement learning-based policies and heuristic greedy policies.

**Notation.** Let $b_t(r, \tau)$ denote the belief matrix at time $t$, and let bs denote the full sequence of beliefs across time, i.e., a 3D array of shape $(T \times 8 \times 20)$. Alternatively, for greedy policies, we use $t_p^{(t)}$, the 8-dimensional vector of last-visit times (for each port $p$ at trial $t$), and $t_p^{(1:T)}$ is the sequence of these vectors over time.

**Input.** Let `data` be the array of experimental pokes from a session, with shape $(T,)$ and integer entries in $\{1, \ldots, 8\}$.

**Procedure.** For each choice of memory parameters $\theta$ and model policy parameter $\varepsilon$, do the following:

1. **Construct initial beliefs:** compute the prior $f_\theta$.
2. **Train policy (only for RL models):** train and freeze a policy using reinforcement learning (e.g., DQN) based on the belief MDP. This step is skipped for greedy models.
3. **Compute belief sequence:** for each experimental session, use the poke history to compute the belief trajectory bs (or the last-visit sequence $t_p^{(1:T)}$) using the belief update rules, starting from $f_\theta$.
4. **Compute action probabilities:** for each timestep $t$, apply the policy (parameterized by $\varepsilon$) to the belief $b_t$ (or $t_p^{(t)}$), yielding a vector $P_t \in \mathbb{R}^8$ of action probabilities. Stack these to form $P \in \mathbb{R}^{T \times 8}$.
5. **Compute log-likelihood:** for the given session, compute the model's log-probability of the actual poke sequence:
$$\mathrm{LL} = \sum_{t=1}^{T} \log P_{t,\mathtt{data}[t]}$$
or in NumPy:
```
LL = np.sum(np.log(P[np.arange(P.shape[0]), data]))
```

Repeat this across all sessions and sum the log-likelihoods to evaluate the model fit.

## 5 Poke count model (aka Gonzalo's model)

I propose the following modification of Gonzalo's model. There are two mechanisms that can potentially stop the trial after a poke (which I will call the "current poke"[3] below), based on

- **Mechanism 1** is triggered when our current poke (i.e. most recent poke) was done in "exploitation mode", which I define as poking the best port, $p_\star$, according to current beliefs and

[3]Even though "the most recent past poke" is perhaps a more accurate name.

the greedy policy (thus $p_\star$ is the arg max port in Eq. (36)).[4] In this definition exploitation vs. exploration can be directly read from the observed data (which gives us $p$, the actually poked port) and the model's prediction of $p_\star$ from Eq. (36). In short, this mechanism is triggered if

$$p = p_\star \tag{46}$$

(So Gonzalo, note that the only change I have made to your mechanism is to replace $p_{\text{correct}}$ with $p_\star$ as given by Eq. (36).

- **Mechanism 2:** whether or not the "current time", i.e. time after the current poke, is above or below a threshold parameters, $t_{\max}$. We define current time as a sum over the past pokes in the trial (including the current poke) of (1) distance travelled to a poked port, $\Delta x_i$ (from the previous poke or from the inital position in the trial), (2) time spent in the port. Thus if the current poke is the $n$-th poke in the trial, this mechanism is triggered if

$$t_n := \sum_{i=1}^{n} (\alpha \Delta x_i + \Delta t) > t_{\max} \tag{47}$$

and this trial ends after the $n$-th poke. Here $\Delta t$ is the time spent in port in each poke (we take this to be a constant for simplicity).

Alternatively, we can make this part parameter-free by defining

$$t_n := \tau_n - t_{\text{start}} > t_{\max} \tag{48}$$

where $\tau_n$ is the time-stamp for the $n$-th poke, and $t_{\text{start}}$ is the timestamp of tone onset (or the start of trial, if we use another definition of trial start). So in this version we only have one parameter: $t_{\max}$.

The trial ends if one or the other mechanism is triggered. We can also make this triggering stochastic: when the first (second) mechanism's condition is satisfied it is activated with probability $q_1$ ($q_2$), and we assume $q_1, q_2 \approx 1$; the probability that mechanism $i$ is triggered is thus $q_i \mathbf{1}(\text{condition}_i)$, where $\text{condition}_1$ is $p_n = p_\star^n$ and $\text{condition}_2$ is $t_n > t_{\max}$. The probability of not terminating the trial after the $n$-th poke is the probability that neither mechanism has activated. Assuming the stochasticity of these two mechanisms is statistically independent, this is given by

$$1 - P_n(\text{termination}) = (1 - q_1 \, \mathbf{1}(p_n = p_\star^n)) \, (1 - q_2 \, \mathbf{1}(t_n > t_{\max})) \tag{49}$$

or

$$\log(1 - P_n(\text{termination})) = \log(1 - q_1 \, \mathbf{1}(p_n = p_\star^n)) + \log(1 - q_2 \, \mathbf{1}(t_n > t_{\max})) \tag{50}$$

[4] Justification: our model's policy is an $\epsilon$-greedy version of Eq. (36), i.e. the policy chooses $p_\star$ with probability $(1 - \epsilon)$, and otherwise chooses a port uniformly at random. It can be shown that, mathematically, we can re-express this as choosing $p_\star$ with propability $\varepsilon = \frac{7}{8}\epsilon$ and otherwise choosing a port uniformly at random from *among the other 7 ports*:

$$\pi(p) = (1 - \epsilon)\delta(p = p_\star) + \epsilon \frac{1}{8} \tag{44}$$

$$= (1 - \varepsilon)\delta(p = p_\star) + \varepsilon(1 - \delta(p = p_\star)) \frac{1}{7} \tag{45}$$

{rewrite}

In choosing $p_\star$ as **exploitation** I'm using Eq. (45) (rather than the first line) as the decomposition of the full policy into a mixture of an exploitation (first term) and exploitation (second term) policies.

The likelihood of observing a trial with $N$ pokes is then the probability that neither mechanism was activated in the pokes 1 to $N-1$, but at least one of the two mechanisms was activated in the $N$-th poke. Thus the log-likelihood for the poke-number component of the model is given by

$$LL_2 = \log P_N(\text{termination}) + \sum_{n=1}^{N-1} \log(1 - P_n(\text{termination})) \tag{51}$$

$$= \log\left[q_1\,\mathbf{1}(p_N = p_N^\star) + q_2\,\mathbf{1}(t_N > t_{\max}) - q_1 q_2 \mathbf{1}(p_N = p_N^\star)\mathbf{1}(t_N > t_{\max})\right] + \sum_{n=1}^{N-1} \log(1 - q_1\,\mathbf{1}(p_n = p_\star^n)) + \sum_{n=1}^{N-1} \log(1 - q_2\,\mathbf{1}(t_n > t_{\max})) \tag{52}$$

$$= \log\left[q_1\,\mathbf{1}(p_N = p_N^\star) + q_2\,\mathbf{1}(t_N > t_{\max}) - q_1 q_2 \mathbf{1}(p_N = p_N^\star)\mathbf{1}(t_N > t_{\max})\right] + \sum_{\substack{n=1 \\ p_n = p_n^\star}}^{N-1} \log(1 - q_1) + \sum_{\substack{n=1 \\ t_n > t_{\max}}}^{N-1} \log(1 - q_2) \tag{53}$$

$$= \log\left[q_1\,\mathbf{1}(p_N = p_N^\star) + q_2\,\mathbf{1}(t_N > t_{\max}) - q_1 q_2 \mathbf{1}(p_N = p_N^\star)\mathbf{1}(t_N > t_{\max})\right] + M_\star \log(1 - q_1) + M_{\text{late}} \log(1 - q_2) \tag{54}$$

where $p_n$ is the actual/observed $n$-th poked port and $p_n^\star$ is the exploitation-only model's prediction (the arg max in Eq. (36)) for the $n$-th poke. In the second form, I defined $M_\star$ ($M_{\text{late}}$) to be the number of exploitative pokes (the number of "late pokes", i.e. pokes for which $t_n > t_{\max}$) among the non-final pokes of the trial (pokes 1 to $N-1$).

**Note:** the full log-likelihdood is the old log-likelihood (for the poke-choice part of the model – the original part) plus $LL_2$.

**Potential trouble**: having made the mechanisms stochastic, prevents this log-likelihood taking minus infinity as value. However, $M_\star$, $p_N^\star$, $M_{\text{late}}$ and $\mathbf{1}(t_N > t_{\max})$ are discrete variables that depend on model parameters, and thus give rise to discontinuous in the log-likelihood (while $M_{\text{late}}$ and $\mathbf{1}(t_N > t_{\max})$ depend on the parameters $\alpha$, $\Delta t$ and $t_{\max}$, the first two discrete variables depend on the parameters of the non-Gonzalo part of the model which determine $p_n^\star$). However, given that we are not using a gradient-based optimization method, but rather the Nedler-Mead algorithm, this is not a major problem. So for now we can ignore this. If this were a real problem, we could easily change mechanism 2 into a more "smooth" mechanism, and maybe a hacky way of changing mechanism 1 (basically turning arg-max to some sort of arg-softmax).

To smooth out the LL we replace

$$\mathbf{1}(t_n > t_{\max}) \longrightarrow \sigma_\beta(t_n - t_{\max}) \tag{55}$$

where I defined

$$\sigma_\beta(x) = \sigma(\beta x) = \frac{1}{1 + e^{-\beta x}} \tag{56}$$

and $\beta$ is large-ish (when $\beta \to \infty$ we obtain the previous expression). For Mechanism we can do two modifications:

1. we call it "exploitation" not just if you poked the maximum-belief port (belief = is your belief on where the correct port is), but if you did that when your were confident where the correct port is. We define the latter condition by $m_n(p_n) > 0.5$ (we could replace the 0.5 with a free parameter again, let's call it $p_{\text{conf}}$), where $m_n(p_n)$, defined in Eq. (33), is the (marginal) belief that the actually poked port, $p_n$, is the correct port.

2. we smooth out the condition $m_n(p_n) > 0.5$.

As a result of these two modifications we replace

$$\mathbf{1}(p_n = p_n^\star) \longrightarrow \sigma_\beta(m_n(p_n) - p_{\text{conf}}) \tag{57}$$

With these replacements the log-likelihood becomes

$$\begin{aligned} LL_2 = \log\left[q_1\,\sigma_{\beta_1}(m_N(p_N) - p_{\text{conf}}) + q_2\,\sigma_{\beta_2}(t_N - t_{\max}) - q_1 q_2 \sigma_{\beta_1}(m_N(p_N) - p_{\text{conf}})\sigma_{\beta_2}(t_N - t_{\max})\right] \\ + \sum_{n=1}^{N-1} \log(1 - q_1 \sigma_{\beta_1}(m_n(p_n) - p_{\text{conf}})) + \sum_{n=1}^{N-1} \log(1 - q_2 \sigma_{\beta_2}(t_n - t_{\max})) \end{aligned} \tag{58}$$

where now I allow for the $\beta$'s of the two mechanisms to be potentially different. Note that this expression is a function of the parameters of the choice-part of the model, $\boldsymbol{\theta}$ (the $\kappa$'s, $w$'s, etc), via the beliefs $m_n(p_n)$, as well as the new parameters $t_{\max}$, $q_1$ and $q_2$. Set $\beta$ to 10. Set $p_{\text{conf}}$ to 0.5 for starters, but it, and $\beta$, can be free/fit parameters.